MAD

LAB 1

Mainactivity.java

```
package com.example.layoutmanager;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   EditText etName, etAge;
   Button btnSubmit;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      etName = findViewById(R.id.etName);
      etAge = findViewById(R.id.etAge);
      btnSubmit = findViewById(R.id.btnSubmit);

      btnSubmit.setOnClickListener(new View.OnClickListener() {
         @Override
         public void onClick(View view) {
            String name = etName.getText().toString();

            Toast.makeText(MainActivity.this,
                  "Welcome " + name,
                  Toast.LENGTH_SHORT).show();
         }
      });
   }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="20dp">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Student Registration"
      android:textSize="24sp"
      android:textStyle="bold"/>

   <EditText
      android:id="@+id/etName"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Name"/>

   <EditText
      android:id="@+id/etAge"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Age"
      android:inputType="number"/>

   <Button
      android:id="@+id/btnSubmit"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Submit"/>

</LinearLayout>
```

LAB 2

Mainactivity.java

```
package com.example.eventlisteners;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnClick;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnClick = findViewById(R.id.btnClick);

        btnClick.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Toast.makeText(
                        MainActivity.this,
                        "Button Clicked!",
                        Toast.LENGTH_SHORT
                ).show();
            }
        });
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center">

    <Button
```

```
    android:id="@+id/btnClick"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Click Me" />

</LinearLayout>
```

LAB 3

Mainactivity.java

```
package com.example.adapters_toast;


import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {

  ListView listView;

  String[] fruits = {
        "Jackfruit",
        "Banana",
        "Grapes",
        "Apple",
        "Mango"
  };

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    listView = findViewById(R.id.listView);

    ArrayAdapter<String> adapter = new ArrayAdapter<>(
        this,
        android.R.layout.simple_list_item_1,
        fruits
    );

    listView.setAdapter(adapter);
```

```
        listView.setOnItemClickListener((parent, view, position, id) -> {

            String selectedFruit = fruits[position];

            Toast.makeText(
                    MainActivity.this,
                    "You selected: " + selectedFruit,
                    Toast.LENGTH_SHORT
            ).show();
        });
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Lists of Fruits"
        android:textSize="24sp"
        android:textStyle="bold"/>

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```

LAB 4

Mainactivity.java

```
package com.studentdatabaseapp;
```

```
import android.database.Cursor;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText editName, editRegister, editDepartment;
    Button buttonSave, buttonDelete;
    TextView textResult;
    DatabaseHelper databaseHelper;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editName = findViewById(R.id.editName);
        editRegister = findViewById(R.id.editRegister);
        editDepartment = findViewById(R.id.editDepartment);
        buttonSave = findViewById(R.id.buttonSave);
        buttonDelete = findViewById(R.id.buttonDelete);
        textResult = findViewById(R.id.textResult);

        databaseHelper = new DatabaseHelper(this);

        displayStudents();

        // SAVE STUDENT
        buttonSave.setOnClickListener(v -> {
```

```
    String name = editName.getText().toString().trim();
    String reg = editRegister.getText().toString().trim();
    String dept = editDepartment.getText().toString().trim();

    if (name.isEmpty() || reg.isEmpty() || dept.isEmpty()) {
        Toast.makeText(this, "Enter all details",
                Toast.LENGTH_SHORT).show();
    } else if (databaseHelper.insertStudent(name, reg, dept)) {

        Toast.makeText(this, "Student Saved",
                Toast.LENGTH_SHORT).show();

        editName.setText("");
        editRegister.setText("");
        editDepartment.setText("");

        displayStudents();

    } else {
        Toast.makeText(this, "Save Failed",
                Toast.LENGTH_SHORT).show();
    }
});

// DELETE STUDENT
buttonDelete.setOnClickListener(v -> {

    String reg = editRegister.getText().toString().trim();

    if (reg.isEmpty()) {
        Toast.makeText(this, "Enter Register Number",
```

```
                Toast.LENGTH_SHORT).show();
        } else if (databaseHelper.deleteStudent(reg)) {

            Toast.makeText(this, "Student Deleted",
                    Toast.LENGTH_SHORT).show();

            editRegister.setText("");
            displayStudents();

        } else {
            Toast.makeText(this, "Student Not Found",
                    Toast.LENGTH_SHORT).show();
        }
    });
}

// DISPLAY STUDENTS AUTOMATICALLY
private void displayStudents() {

    Cursor c = databaseHelper.getAllStudents();
    StringBuilder result = new StringBuilder();

    while (c.moveToNext()) {
        result.append("ID: ").append(c.getInt(0))
                .append("\nName: ").append(c.getString(1))
                .append("\nRegister No: ").append(c.getString(2))
                .append("\nDepartment: ").append(c.getString(3))
                .append("\n----------------------\n");
    }

    c.close();
```

```
        textResult.setText(result.length() == 0 ?
                "No students found." : result.toString());
    }
}
```

DataBaseHelper.java

```
package com.example.lab4_studentdatabaseapp;

import android.content.*;
import android.database.*;
import android.database.sqlite.*;

public class DatabaseHelper extends SQLiteOpenHelper {

    public DatabaseHelper(Context context) {
        super(context, "StudentDB.db", null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE students (id INTEGER PRIMARY KEY, " +
                "name TEXT, register_no TEXT, department TEXT)");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS students");
        onCreate(db);
    }

    // Insert Student with smallest available ID
    public boolean insertStudent(String name, String reg, String dept) {
        SQLiteDatabase db = getWritableDatabase();

        int id = 1;
        Cursor c = db.rawQuery("SELECT id FROM students ORDER BY id", null);

        while (c.moveToNext()) {
            if (c.getInt(0) == id) id++;
            else if (c.getInt(0) > id) break;
        }
        c.close();
```

```
        ContentValues values = new ContentValues();
        values.put("id", id);
        values.put("name", name);
        values.put("register_no", reg);
        values.put("department", dept);

        return db.insert("students", null, values) != -1;
    }

    // View All Students
    public Cursor getAllStudents() {
        return getReadableDatabase().rawQuery(
                "SELECT * FROM students ORDER BY id", null);
    }

    // Delete Student
    public boolean deleteStudent(String reg) {
        return getWritableDatabase().delete(
                "students", "register_no=?", new String[]{reg}) > 0;
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Student Database"
        android:textSize="26sp"
        android:textStyle="bold"
        android:gravity="center"
        android:padding="10dp"/>

    <EditText
        android:id="@+id/editName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
```

```
    android:hint="Student Name"
    android:inputType="textPersonName"/>

<EditText
    android:id="@+id/editRegister"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Register Number"
    android:inputType="text"/>

<EditText
    android:id="@+id/editDepartment"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Department"
    android:inputType="text"/>

<Button
    android:id="@+id/buttonSave"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="SAVE STUDENT"
    android:layout_marginTop="15dp"/>

<Button
    android:id="@+id/buttonDelete"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="DELETE STUDENT"/>

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Saved Student Details"
    android:textSize="20sp"
    android:textStyle="bold"
    android:layout_marginTop="15dp"/>

<ScrollView
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1">

    <TextView
        android:id="@+id/textResult"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="No students found."
```

```
        android:textSize="16sp"
        android:padding="10dp"/>

    </ScrollView>

</LinearLayout>
```

LAB 5

1. AndroidManifest.xml

Add Internet permission above the <application> tag:

<uses-permission android:name="android.permission.INTERNET"/>

Mainactivity.java

```
package com.example.rssfeedreader;


import android.os.Bundle;

import android.widget.*;

import android.view.View;

import android.content.Intent;

import android.net.Uri;


import androidx.appcompat.app.AppCompatActivity;


import org.xmlpull.v1.XmlPullParser;

import android.util.Xml;


import java.net.URL;

import java.net.HttpURLConnection;

import java.util.ArrayList;

import java.util.concurrent.Executors;
```

```
public class MainActivity extends AppCompatActivity {

    Button btnLoad;
    ListView listView;

    ArrayList<String> titles = new ArrayList<>();
    ArrayList<String> links = new ArrayList<>();

    ArrayAdapter<String> adapter;

    String RSS_URL = "https://feeds.bbci.co.uk/news/rss.xml";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnLoad = findViewById(R.id.btnLoad);
        listView = findViewById(R.id.listView);

        adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                titles);

        listView.setAdapter(adapter);

        btnLoad.setOnClickListener(v -> loadRSS());

        listView.setOnItemClickListener((p, v, pos, id) -> {
            startActivity(new Intent(Intent.ACTION_VIEW,
```

```
            Uri.parse(links.get(pos))));
    });
}

void loadRSS() {

    btnLoad.setEnabled(false);

    Executors.newSingleThreadExecutor().execute(() -> {

        ArrayList<String> newTitles = new ArrayList<>();
        ArrayList<String> newLinks = new ArrayList<>();

        try {
            HttpURLConnection con =
                    (HttpURLConnection) new URL(RSS_URL).openConnection();

            con.setConnectTimeout(10000);
            con.setReadTimeout(10000);

            XmlPullParser parser = Xml.newPullParser();
            parser.setInput(con.getInputStream(), null);

            String title = "", link = "";
            int event = parser.getEventType();

            while (event != XmlPullParser.END_DOCUMENT) {

                if (event == XmlPullParser.START_TAG) {

                    String tag = parser.getName();
```

```
        if (tag.equals("item")) {
            title = "";
            link = "";
        }

        else if (tag.equals("title") &&
                title.isEmpty()) {
            title = parser.nextText();
        }

        else if (tag.equals("link") &&
                link.isEmpty()) {
            link = parser.nextText();
        }
    }

    else if (event == XmlPullParser.END_TAG &&
            parser.getName().equals("item")) {

        if (!title.isEmpty() && !link.isEmpty()) {
            newTitles.add(title);
            newLinks.add(link);
        }
    }

    event = parser.next();
}

con.disconnect();
```

```
        } catch (Exception e) {
            e.printStackTrace();
        }

        runOnUiThread(() -> {

            titles.clear();
            links.clear();

            titles.addAll(newTitles);
            links.addAll(newLinks);

            adapter.notifyDataSetChanged();

            btnLoad.setEnabled(true);

            Toast.makeText(this,
                    titles.isEmpty() ? "Failed to load RSS feed"
                            : "RSS Feed Loaded",
                    Toast.LENGTH_SHORT).show();
        });
    });
  }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
```

```
    android:orientation="vertical"
    android:padding="10dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="RSS FEED READER"
        android:textSize="24sp"
        android:textStyle="bold"
        android:gravity="center"/>

    <Button
        android:id="@+id/btnLoad"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Load RSS Feed"/>

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>

</LinearLayout>
```

LAB 6

Mainactivity.java

```
package com.example.multithread;

import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
```

```
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    TextView textResult;
    Button btnStart, btnStop;

    Thread thread;
    volatile boolean running = false;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        textResult = findViewById(R.id.textResult);
        btnStart = findViewById(R.id.btnStart);
        btnStop = findViewById(R.id.btnStop);

        btnStart.setOnClickListener(v -> {

            if (running) {
                return;
            }

            running = true;

            thread = new Thread(() -> {

                for (int i = 1; i <= 10; i++) {

                    if (!running) {
                        break;
                    }

                    final int count = i;

                    runOnUiThread(() -> {
                        textResult.setText("Count: " + count);
                    });

                    try {
                        Thread.sleep(1000);
                    }
                    catch (InterruptedException e) {
                        Thread.currentThread().interrupt();
                        break;
                    }
```

```
            }

            runOnUiThread(() -> {
                if (running) {
                    textResult.setText("Thread Completed!");
                } else {
                    textResult.setText("Thread Stopped!");
                }
                running = false;
            });

        });

        thread.start();
    });

    btnStop.setOnClickListener(v -> {
        running = false;

        if (thread != null) {
            thread.interrupt();
        }
    });
  }
}
```

activity_main.xml

```
<LinearLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:gravity="center"
  android:padding="20dp">

  <TextView
    android:id="@+id/textTitle"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Multithreading Application"
    android:textSize="24sp"
    android:textStyle="bold"
    android:layout_marginBottom="30dp"/>

  <TextView
    android:id="@+id/textResult"
```

```
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Click Start Thread"
    android:textSize="20sp"
    android:layout_marginBottom="30dp"/>

  <Button
    android:id="@+id/btnStart"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Start Thread"/>

  <Button
    android:id="@+id/btnStop"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Stop Thread"/>

</LinearLayout>
```

LAB 7

Step 1: Add Internet permission

Open AndroidManifest.xml and add this permission before the <application> tag:

```
<uses-permission android:name="android.permission.INTERNET"/>
```

Mainactivity.java

```
package com.example.gpslocation;

import android.location.Address;
import android.location.Geocoder;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;
import java.io.IOException;
import java.util.List;
import java.util.Locale;

public class MainActivity extends AppCompatActivity {

  EditText edtPlace;
```

```
Button btnSearch;
TextView txtLocation;

@Override
protected void onCreate(Bundle savedInstanceState) {
   super.onCreate(savedInstanceState);
   setContentView(R.layout.activity_main);

   edtPlace = findViewById(R.id.edtPlace);
   btnSearch = findViewById(R.id.btnSearch);
   txtLocation = findViewById(R.id.txtLocation);

   btnSearch.setOnClickListener(v -> searchLocation());
}

void searchLocation() {
   String place = edtPlace.getText().toString().trim();

   if (place.isEmpty()) {
      edtPlace.setError("Enter a place");
      return;
   }

   txtLocation.setText("Searching...");

   new Thread(() -> {
      try {
         Geocoder geocoder = new Geocoder(this, Locale.getDefault());
         List<Address> addresses =
               geocoder.getFromLocationName(place, 1);

         runOnUiThread(() -> {
            if (addresses != null && !addresses.isEmpty()) {
               Address address = addresses.get(0);

               txtLocation.setText(
                     "Place: " + place +
                           "\nLatitude: " + address.getLatitude() +
                           "\nLongitude: " + address.getLongitude()
               );
            } else {
               txtLocation.setText("Place not found!");
            }
         });

      } catch (IOException e) {
         runOnUiThread(() ->
               txtLocation.setText("Unable to get location. Check internet."));
```

```
        }
      }).start();
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/edtPlace"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter place name"/>

    <Button
        android:id="@+id/btnSearch"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Search Location"/>

    <TextView
        android:id="@+id/txtLocation"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Location details appear here"
        android:textSize="18sp"
        android:padding="20dp"/>

</LinearLayout>
```

LAB 8

Step 1: Add Internet permission

Open AndroidManifest.xml and add this permission before the <application> tag:

```
<uses-permission android:name="android.permission.INTERNET"/>
```

Mainactivity.java

```
package com.example.gpsstorage;

import android.content.Intent;
import android.net.Uri;
import android.location.*;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;
import java.io.OutputStream;
import java.io.IOException;
import java.util.*;

public class MainActivity extends AppCompatActivity {

   EditText edtPlace;
   Button btnSearch;
   TextView txtLocation;
   String data;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      edtPlace = findViewById(R.id.edtPlace);
      btnSearch = findViewById(R.id.btnSearch);
      txtLocation = findViewById(R.id.txtLocation);

      btnSearch.setOnClickListener(v -> searchLocation());
   }

   void searchLocation() {
      String place = edtPlace.getText().toString().trim();

      if (place.isEmpty()) {
         edtPlace.setError("Enter place name");
         return;
      }

      new Thread(() -> {
         try {
            Geocoder geocoder = new Geocoder(this, Locale.getDefault());
```

```
            List<Address> list =
                  geocoder.getFromLocationName(place, 1);

            runOnUiThread(() -> {
               if (list != null && !list.isEmpty()) {
                  Address a = list.get(0);

                  data = "Place: " + place +
                        "\nLatitude: " + a.getLatitude() +
                        "\nLongitude: " + a.getLongitude();

                  txtLocation.setText(data);
                  saveFile();
               } else {
                  txtLocation.setText("Place not found!");
               }
            });

         } catch (IOException e) {
            runOnUiThread(() ->
                  txtLocation.setText("Location search failed"));
         }
      }).start();
   }

   void saveFile() {
      Intent intent = new Intent(Intent.ACTION_CREATE_DOCUMENT);
      intent.setType("text/plain");
      intent.addCategory(Intent.CATEGORY_OPENABLE);
      intent.putExtra(Intent.EXTRA_TITLE, "Location.txt");
      startActivityForResult(intent, 1);
   }

   @Override
   protected void onActivityResult(int requestCode,
                           int resultCode, Intent result) {
      super.onActivityResult(requestCode, resultCode, result);

      if (requestCode == 1 && resultCode == RESULT_OK
            && result != null) {
         Uri uri = result.getData();

         try {
            OutputStream output =
                  getContentResolver().openOutputStream(uri);

            if (output != null) {
               output.write(data.getBytes("UTF-8"));
```

```
                output.close();
                Toast.makeText(this, "Saved successfully!",
                        Toast.LENGTH_SHORT).show();
            }

        } catch (IOException e) {
            Toast.makeText(this, "File saving failed",
                    Toast.LENGTH_SHORT).show();
        }
      }
   }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">

   <EditText
      android:id="@+id/edtPlace"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter place name"/>

   <Button
      android:id="@+id/btnSearch"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Search and Save"/>

   <TextView
      android:id="@+id/txtLocation"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Location details"
      android:textSize="18sp"
      android:padding="20dp"/>

</LinearLayout>
```

LAB 9

Step 1: Add permissions

Open AndroidManifest.xml and add these permissions before the <application> tag:

<uses-permission android:name="android.permission.SEND_SMS"/>

Mainactivity.java

```
package com.example.smsalert;

import android.Manifest;
import android.content.pm.PackageManager;
import android.os.Bundle;
import android.telephony.SmsManager;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   EditText number, message;
   Button send;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      number = findViewById(R.id.number);
      message = findViewById(R.id.message);
      send = findViewById(R.id.send);

      send.setOnClickListener(v -> {
         if (number.getText().toString().isEmpty() ||
               message.getText().toString().isEmpty()) {
            Toast.makeText(this, "Enter all details",
                  Toast.LENGTH_SHORT).show();
         } else if (checkSelfPermission(Manifest.permission.SEND_SMS)
               != PackageManager.PERMISSION_GRANTED) {
            requestPermissions(
                  new String[]{Manifest.permission.SEND_SMS}, 1);
         } else {
            sendSMS();
         }
      });
   }
```

```
    void sendSMS() {
        try {
            SmsManager sms = SmsManager.getDefault();
            sms.sendTextMessage(
                    number.getText().toString(),
                    null,
                    message.getText().toString(),
                    null, null);

            Toast.makeText(this, "Message Sent Successfully",
                    Toast.LENGTH_SHORT).show();

        } catch (Exception e) {
            Toast.makeText(this, "Message Sending Failed",
                    Toast.LENGTH_SHORT).show();
        }
    }

    @Override
    public void onRequestPermissionsResult(int requestCode,
                                           String[] permissions, int[] results) {
        super.onRequestPermissionsResult(requestCode, permissions, results);

        if (requestCode == 1 && results.length > 0 &&
                results[0] == PackageManager.PERMISSION_GRANTED) {
            sendSMS();
        }
    }
}
```

activity_main.xml

```
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center">

    <EditText
        android:id="@+id/number"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Mobile Number"
        android:inputType="phone"/>
```

```
    <EditText
        android:id="@+id/message"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Alert Message"/>

    <Button
        android:id="@+id/send"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send SMS"/>

</LinearLayout>
```

LAB 10

Mainactivity.java

```
package com.example.gameapp;

import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;
import java.util.Random;

public class MainActivity extends AppCompatActivity {

    EditText guess;
    TextView result;
    int number;

    void newGame() {
        number = new Random().nextInt(10) + 1;
        result.setText("Guess a number!");
        guess.setText("");
    }

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        guess = findViewById(R.id.guess);
        result = findViewById(R.id.result);
        Button check = findViewById(R.id.check);
```

```
        Button restart = findViewById(R.id.restart);

        newGame();

        check.setOnClickListener(v -> {
            String s = guess.getText().toString();

            if (s.isEmpty()) {
                guess.setError("Enter a number");
                return;
            }

            int n = Integer.parseInt(s);

            if (n < 1 || n > 10) {
                result.setText("Enter between 1 and 10");
            } else if (n == number) {
                result.setText("Congratulations! You Win!");
            } else if (n < number) {
                result.setText("Too Low! Try Again");
            } else {
                result.setText("Too High! Try Again");
            }
        });

        restart.setOnClickListener(v -> newGame());
    }
}
```

activity_main.xml

```
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Guess the Number (1-10)"
        android:textSize="22sp"/>

    <EditText
        android:id="@+id/guess"
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter your guess"
        android:inputType="number"/>

    <Button
        android:id="@+id/check"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Check"/>

    <TextView
        android:id="@+id/result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp"
        android:padding="15dp"/>

    <Button
        android:id="@+id/restart"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Restart Game"/>

</LinearLayout>
```